

9 Time

Tune In

- a. Eggs (bottom centre box); caterpillar (middle left box); chrysalis (Top box); Butterfly (middle right box).
- b. First stage: eggs; second stage: caterpillar; third stage: chrysalis; fourth stage: Adult butterfly.

Exercise 9A

- 2.a. 3 : 17 b. 11 : 48 c. 7 : 10 d. 1 : 00 e. 6 : 32
- 3.a. 2 : 30 b. 3 : 49 c. 4 : 11 d. 8 : 22 e. 11 : 53

Application based question

4.



Let's Try! 1

- 1.a. 3: 35 am = 03: 35 hours b. 4:25 pm = 16:25 hours
- 2.a. 05:35 hours = 05: 35 am d. 17 : 50 hours = 5:50 pm

Exercise 9B

- 3.a. 7:10 pm = 19:10 hrs b. 10:55 am = 10:55 hrs c. 12.05 am = 00:05 hrs
d. 12:30 pm = 12:30 hrs e. 6:40 am = 06:40 hrs f. 1: 22 pm = 13:22 hours
g. 9:00 pm = 21:00 hrs h. 11:15 am = 11:15 hrs
- 4.a. 05:45 hrs = 5:45 am b. 22:17 hrs = 10:17 pm c. 00:10 hrs = 12:10 am
d. 16:22 hrs = 4:22 pm e. 12:28 hrs = 12:28 pm f. 13:45 hrs = 1:45 pm
g. 09:24 hrs = 9:24 am h. 02:05 hrs = 2:05 am

5.

	Time in words	24-h clock	12-h clock
a.	7 o'clock in the evening	19:00 hrs	7:00 pm
b.	2 hours 20 minutes in the afternoon	14:20 hrs	2:20 pm
c.	12 noon or Midday	12:00 hrs	12:00 pm
d.	7 hours 10 minutes 10 in the evening	19:10 hrs	7:10 pm
e.	midnight	00:00 hrs	12 midnight
f.	Two hours 45 minutes at night	02:45 hrs	2:45 am
g.	Twenty minutes to two in the afternoon	13:40 hrs	1:40 pm
h.	17 minutes past four in the morning	04:17 hrs	4:17 am
i.	Quarter past ten in the evening	22:15 hrs	10:15 pm

6.a. The play started at 18:30 hours.

b. Mr Roy goes to office at 08:45 hours.

c. Raghav comes from school at 14:10 hours.

d. The school bus comes at 07:10 hours.

e. Riya eats her dinner at 20:15 hours.

7. Stephan arrived first as he came at 15.45 hours or 3:45 pm in the evening almost an hour before John, who came at 4:47 pm.

Let's Try! 2

1. 2.20 am to 3.15 am

$$2:20 \text{ am} + \underline{40 \text{ minutes}} = 3:00 \text{ am} + \underline{15 \text{ minutes}} = 3:15 \text{ am}$$

$$\text{Duration} = 40 + 15 \text{ minutes}$$

$$= 55 \text{ minutes (Ans)}$$

2. Duration between 11:30 am and 5:40 pm

$$11:30 + \underline{30 \text{ mins}} = 12:00 \text{ pm} + \underline{5 \text{ hours}} = 5:00 \text{ pm} + \underline{40 \text{ mins}} = 5:40 \text{ pm}$$

$$\text{Duration} = 30 \text{ mins} + 5 \text{ hrs} + 40 \text{ mins}$$

$$= 5 \text{ hrs} + 70 \text{ mins}$$

$$= 5 \text{ hrs} + 1 \text{ hour} + 10 \text{ mins} = 6 \text{ hrs } 10 \text{ mins (Ans)}$$

Exercise 9C

2.a. 10:10 am to 11:00 am

$$10:10 + \underline{50 \text{ mins}} = 11:00 \text{ am} \quad =$$

$$\text{Duration} = 50 \text{ mins. (Ans)}$$

b. 8:15 am to 2:30 pm

$$8:15 + \underline{45 \text{ mins}} = 9:00 \text{ am} + \underline{5 \text{ hrs}} = 2:00 \text{ pm} + \underline{30 \text{ mins}} = 2:30 \text{ pm}$$

$$\text{Duration} = 45 \text{ mins} + 5 \text{ hrs} + 30 \text{ mins}$$

$$\begin{aligned}
 &= 5 \text{ hrs} + 75 \text{ mins} \\
 &= 5 \text{ hrs} + 1 \text{ hour} + 15 \text{ mins} \\
 &= 6 \text{ hours} + 15 \text{ mins} \text{ (Ans)}
 \end{aligned}$$

c. 5:20 pm to 9:50 pm

$$5:20\text{pm} + \underline{40 \text{ mins}} = 6:00 \text{ pm} + \underline{3 \text{ hrs}} = 9:00 \text{ pm} + \underline{50 \text{ mins}} = 9:50 \text{ pm}$$

$$\text{Duration} = 40 \text{ mins} + 3 \text{ hours} + 50 \text{ mins}$$

$$= 3 \text{ hrs} + 90 \text{ mins}$$

$$= 4 \text{ hrs} + 30 \text{ mins} \text{ (Ans)}$$

d. 9:35 pm to 1.20 am

$$9:35 \text{ pm} + \underline{25 \text{ mins}} = 10 \text{ pm} + \underline{3 \text{ hrs}} = 1:00 \text{ am} + \underline{20 \text{ mins}} = 1.20 \text{ am}$$

$$\text{Duration} = 25 \text{ mins} + 3 \text{ hrs} + 20 \text{ mins}$$

$$= 3 \text{ hrs} 45 \text{ mins} \text{ (Ans)}$$

3.a. 6.45 am + 15 mins = 7:00 am + 10 mins = 7:10 am

Ans: 7:10 am

b. 50 mins after 2:50 pm

$$2:50 \text{ pm} + \underline{10 \text{ mins}} = 3:00 \text{ pm} + \underline{40 \text{ mins}} = 3:40 \text{ pm}$$

Ans: 3:40 pm

c. 1 hour after 9:20 am = 10: 20 am (Ans)

d. 10.30 am + 4 hours = 2:30 pm + 15 mins = 2:45 pm (Ans)

Application Based Questions

4. Starting time of concert - 5:15 pm

Finishing time - 9:00 pm

Duration of concert

$$= 5:15 \text{ pm} + \underline{45 \text{ mins}} = 6:00 \text{ pm} + \underline{3 \text{ hrs}} = 9:00 \text{ pm}$$

$$\text{duration: } 45 \text{ mins} + 3 \text{ hrs}$$

Ans: The duration of the violin concert was 3 hours 45 mins.

5. Flight take-off time- 04:45 hrs

Landing time - 09:15 hrs

$$\text{Duration} = 04:45 + \underline{15 \text{ mins}} = 05:00 \text{ hrs} + \underline{4 \text{ hrs}} = 09:00 \text{ hrs} + \underline{15 \text{ mins}} = 09:15 \text{ hours}$$

$$= 15 \text{ mins} + 4 \text{ hrs} + 15 \text{ mins}$$

$$= 4\text{hrs} 30 \text{ mins}$$

Ans: The duration of the Chennai Dubai flight was 4 hours 30 mins.

6. Beginning time of show= 8:30 pm

Ending time of show 9:15 pm

$$8:30 \text{ pm} + \underline{30 \text{ mins}} = 9:00 \text{ pm} + \underline{15 \text{ mins}} = 9:15 \text{ pm}$$

$$\text{Duration of the show} = 30 \text{ mins} + 15 \text{ mins}$$

$$= 45 \text{ mins}$$

Ans: The duration of the educational show is 45 minutes,

7. Gopal takes 50 minutes + 10 minutes to help cook and clean up the kitchen.

Thus, he takes 60 minutes or one hour to help out.

If he started helping at 10:20 am, he finished his work at 11:20 am after an hour.

Ans: Gopal finished his work at 11:20 am.

8. Train departure time 6:45 am

Duration = 4 hrs 20 mins

Arrival time in Jaipur: 06:45 hrs + 4 hrs 20 mins

$$06:45 + \underline{4 \text{ hrs}} = 10:45 \text{ hrs} + \underline{15 \text{ mins}} = 11:00 \text{ hrs} + \underline{5 \text{ mins}} = 11:05 \text{ hrs}$$

Ans: The train arrived in Jaipur at 11:05 am.

9. Time shown on Smart watch = 22:45 hrs

Actual time = clock time + 20 mins

$$= 22:45 : \underline{15 \text{ mins}} = 23:00 \text{ hrs} + \underline{5 \text{ mins}} = 23:05 \text{ hrs}$$

Actual time is 23:05 hours or 11:05 pm

Ans: The correct time is 11:05 pm.

10. Starting time of Annual Day function: 10:30 am

Duration : 2 hours 40 mins

Finishing time = Starting time + duration

$$= 10.30 \text{ am} + \underline{2 \text{ hours}} = 12:30 \text{ pm} + \underline{30 \text{ mins}} = 1:00 \text{ pm} + \underline{10 \text{ mins}} = 1:10 \text{ pm}$$

Ans: The annual day show ended at 1:10 pm.

Exercise 9D

- 2.a. Number of days from March 15 to April 13

= Number of days in March + Number of days in April

= 31-14 days of March + 13 days of April

= 17 + 13 days

= 30 days (**Ans**)

- b. From 12th Nov. to 12th December

= Number of days of Nov. + Number of days of December

= 30 – 11 days of November + 12 days of December

= 19 + 12

= 31 days (**Ans**)

c. From 8th August to 18th October

= Number of days of August + Number of days of September + Number of days of Oct.

= 31-7 days + 30 days + 10 days

= 24 + 30 + 18

= 72 days **(Ans)**

d. From 20th April to 1st June

= Number of days of April + Number of days of May + 1st of June

= 30 – 19 + 31 + 1

= 11 + 31 + 1

= 43 days **(Ans)**

Application Based Questions

3.a. Eating an ice-cream takes minutes.

b. Visiting a beach – takes hours

c. Painting a house takes days

d. Buying tickets : takes minutes

e. Doing homework : takes hours

f. Stitching clothes takes hours.

4. No. of days Bhanu was on leave = No. of leave days in August + No. of leave days in Sept.

= (31-25) days in August + 12 days of September

= 6 + 12 = 18 days

Ans: Bhanu was on leave for 18 days.

5. Duration of saree weaving: No. of weaving days in June + No. of weaving days in July

= 30 + 10

= 40 days.

Ans: The weaver thus took 40 days to weave the saree.

6. No. of summer camp days = No of camp days in May + Number of camp days in June

31-6 days of May + 15 days of June

= 25 + 15 days

= 40 days

Ans: The summer camp lasted 40 days.

7. Winter vacation days = No. of vacation days in Dec. + no. of vacation days in Jan.

= 31 – 24 + 15 days of January

= 7 + 15 days

= 22 days

Ans: Winter vacations of the school lasted 22 days.

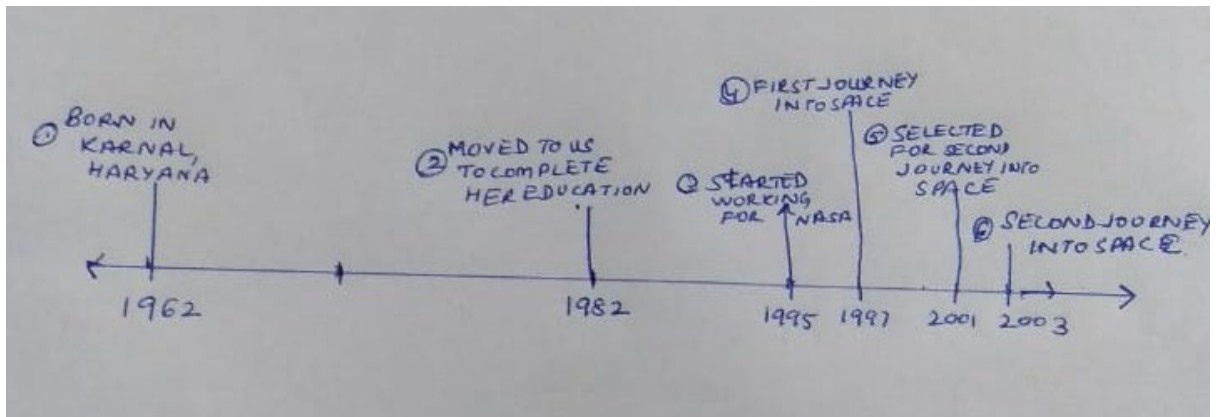
8. Three leap years after 2020 will be

2024, 2028 and 2032.

Exercise 9E

1. The important events in Kalpana Chawla's life marking her progress are as follows

1. 1962: Born in Karnal, Haryana.
2. 1982: Moved to US to complete her education.
3. 1995: Started working with NASA.
4. 1997: First Journey into space.
5. 2001: Selected for second journey into space.
6. 2003: Second journey into space.



Revision Exercises

1.a. The teacher went to school at 7:15 a.m.

b. Rashmi went out for shopping with her friends at 6:30 p.m.

c. Salma's mother packed lunch box for her at 6:45 a.m.

d. Rohan had his dinner with his family at 8:30 p.m.

2.

	Time	12-h clock	24-h clock
a.	10 past six in the morning	6:10 am	06:10 hrs
b.	Quarter to three in the evening	2:45 pm	14:45 hrs
c.	Half past six in the evening	6:30 pm	18:30 hrs
d.	12 noon	12:00 pm	12:00 hrs
e.	Half past 12 in the afternoon	12:30 pm	12:30 hrs
f.	Five past 8 in the evening	8:05 pm	20:05 hrs
g.	Twelve to 6 in the morning	5:48 a.m	05:48 hrs
h.	17 minutes past 4 in the morning	4:17 am	04:17 hrs

3.

	Start Time	End Time	Time Taken
a.	06:00 hrs	21:25 hrs	15:25 hrs
b.	04:30 hrs	08:15 hrs	3 h 45 mins
c.	00:00 hrs	12:00 hrs	12 noon
d.	05:40 hrs	12:00 hrs	6 h 20 mins
e.	01:45 hrs	17:00 hrs	15 h 15 mins
f.	12:00 hrs	02:40 hrs	14 h 40 min
g.	15:15 hrs	23:00 hrs	7 h 45 mins
h.	21:25 hrs	06:00 hrs	8 h 35 mins
i.	04:30 am	11:00 am	6 h 30 mins
j.	6.25 pm	7:10 pm	45 mins
k.	12:30 am	12:00 noon	11 h 30 mins
l.	6:00 am	8:15 am	2 h 15 mins
m.	7:45 am	1:30 pm	5 h 45 mins
n.	12 noon	12 noon	24 h
o.	4.30 pm	2:30 am	10 h
p.	2:30 pm	6:00 pm	3 h 30 mins

Level Up

1. Starting time of project: 10:20 am

working time : 3 h 50 mins

Rohan stopped work at : ?

$$\begin{aligned}
 &10:20 \text{ am} + 3 \text{ h } 50 \text{ mins} \\
 = &10:20 \text{ am} + \underline{40 \text{ mins}} = 11:00 \text{ am} + \underline{3 \text{ hrs}} = 2:00 \text{ pm} + \underline{10 \text{ mins}} = 2:10 \text{ pm}
 \end{aligned}$$

Ans: The time was 2:10 pm when Rohan's mother called him for lunch.

2. Duration of the film: 4:40 pm to 6:20 pm

$$\begin{aligned}
 &: 4:40 + \underline{20 \text{ mins}} = 5:00 \text{ pm} + \underline{1 \text{ h}} = 6:00 \text{ pm} + \underline{20 \text{ mins}} = 6:20 \text{ pm} \\
 = &20 \text{ mins} + 1 \text{ h} + 20 \text{ mins} \\
 = &1 \text{ h } 40 \text{ mins}
 \end{aligned}$$

Kumar and his friends took break = 15 mins.

$$\begin{aligned}
 \text{Duration for which film was watched} &= 1 \text{ h } 40 \text{ mins} - 15 \text{ mins} \\
 &= 1 \text{ h } 25 \text{ mins}
 \end{aligned}$$

Ans: Kumar and his friends watched the film for 1 h 25 mins.

3. Duration of Rubina's zoo visit: 9:25 am to 2:30 pm

$$\begin{aligned}
 &= 9:25 + \underline{35 \text{ mins}} = 10:00 \text{ am} + \underline{4 \text{ hrs}} = 2:00 \text{ pm} + \underline{30 \text{ mins}} = 2:30 \text{ pm} \\
 &= 35 \text{ mins} + 4 \text{ h} + 30 \text{ mins} \\
 &= 5 \text{ h } 05 \text{ mins}
 \end{aligned}$$

Ans: Rubina spent 5 h 5 minutes in the Darjeeling zoo.

4. Train departure from Chennai: 07:30 hrs

Arrival time in Bengaluru: 14:10 hrs

Duration of journey =

$$\begin{aligned} &07:30 \text{ hrs} + \underline{30 \text{ min}} = 0800 \text{ hrs} + \underline{6 \text{ h}} = 14:00 \text{ hrs} + \underline{10 \text{ min}} = 14:10 \text{ hrs} \\ &= 30 \text{ mins} + 6 \text{ h} + 10 \text{ mins} \\ &= 6 \text{ h } 40 \text{ mins} \end{aligned}$$

Ans: The train journey from Chennai to Bengaluru lasted 6 h 40 mins.

5. Duration of Summer vacation = No. of holidays in May + No. of holidays in June
= 31 – 10 days of May + 25 days of June
= 21 days + 25 days
= 46 days

Ans: The school will remain closed for 45 days as summer vacation.

6. Mr Sharma's stay in Agra = No. of days from September + Number of days from October
= 30 - 25 days of September + 3 days of October
= 5 days + 3 days
= 8 days

Ans: Mr Sharma stayed in Agra for 8 days.

7. a. On routine days Rahul comes back from the park 1 h 20 mins after 5.25 pm
= 6:45 pm + 3 mins to get home
= 6:48 pm
b. On the day Rahul rescued the dog he arrived home at 7:50 pm.
Thus, he was late by

$$\begin{aligned} &6:48 + \underline{1 \text{ h}} = 7:48 + \underline{2 \text{ mins}} = 7:50 \\ &= 1 \text{ h } 2 \text{ mins} \end{aligned}$$

Ans: Rahul was late by 1 hour 2 minutes on the day he rescued the dog.

Hots Questions

1. Reflection of the clock will give a laterally inverted image so the correct time should be the flipped image of the clock hands at 4:45 should be 7:15. Thus, the correct time should be 7:15 am.
2. To sleep for nine hours before 6.am, Esther should go to bed at 6:00 am – 9 h
= 6:00 am – 6 hrs = 12 midnight – 3 h = 9:00 pm

Ans: Esther should go to bed at 9:00 pm.

Everyday Maths

1. The shelf-life of biscuits = duration from date of manufacturing to Best Before date
= from May 25 to September 10
= 31- 24 days + June + July + August + 10 days of September
= 7 + 30 + 31 + 31 + 9 days (since it best before date sept 10 not included)
= 108 days

Ans: The shelf life of the biscuits is 108 days.

- b. No, it is not advisable to use the butter after its 'use by' date.